

Jharkhand University of Technology, Ranchi**Diploma 2nd Semester Examination, 2024 (NEP)****Subject : Python Programming****Subject Code : CSE201****Time Allowed : 3 Hours****Full Marks : 70***Answer in your own words.**Answer any five questions.**Question No. 1 is compulsory, and from rest of the questions answer any four only.**All questions carry equal marks.***1. Objective type questions pick the right options:****2×7=14**

- (i) Which of the following is the correct extension for a Python file?
- (a) .pyth (b) .pt
(c) .py (d) .p
- (ii) How do you insert comments in Python code?
- (a) //This is a comment (b) #This is a comment
(c) /*This is a comment*/ (d) --This is a comment
- (iii) What is the output of the following code?
- ```
print(2**3)
```
- (a) 6 (b) 8  
(c) 9 (d) 10
- (iv) Which of the following is used to create a set in Python?
- (a) {} (b) set()  
(c) [] (d) set[ ]
- (v) Which of the following is the correct way to define a list in Python?
- (a) list = [1, 2, 3, 4] (b) list = {1, 2, 3, 4}  
(c) list = (1, 2, 3, 4) (d) list = <1, 2, 3, 4>
- (vi) Which of the following is used to define a block of code in Python?
- (a) Brackets { } (b) Parentheses ()  
(c) Indentation (d) Commas ,
- (vii) Which of the following statements is true about tuples in Python?
- (a) Tuples are mutable  
(b) Tuples are immutable  
(c) Tuples are created using square brackets []  
(d) Tuples can have only one element

2. (a) Define character set in Python. What is Token? Discuss briefly the types of token in Python. 7  
(b) What is variable? Explain the type of variable initialization in python with example.
3. (a) How many types of data structures are there in python? Explain briefly. Explain how Slicing are performed in the python list data structure. 7  
(b) Discuss different flow control statements in python. Explain **if-else** and **if-elif** statement with suitable example. 7
4. (a) Give introduction of **NumPy** module. Explain the **NumPy** statistical functions with example. 7  
(b) What is a **function** in Python? How do you define a function in Python? Write a function in python to print the factorial of the number. 7
5. (a) What is the purpose of the **array** module in Python? How do you create an **array** of integers using the **array** module in Python? 7  
(b) Write a program in python to accept a number from user and check whether it is prime number or not. 7
6. (a) Why modules are used in python? Explain how the modules are created and import in python. 7  
(b) Discuss types of Error in python program. What is Exception Handling? Discuss how to catch exceptions in python. 7
7. Write a short notes on *any four* of the following. 3.5×4=14  
(a) Types of loop  
(b) Indentation  
(c) Comments  
(d) Mutable data type  
(e) Typecasting